

Probability and Statistics

Lecture 1: Basic Concepts of Probability Theory

Topics in this lecture

- Sample space
- Events
- Probability

Why Learn Prob & Stat?

- Give you the power to model things that are random in nature
- Can be applied in many fields of study
 - Science
 - Engineering
 - Business

Motivating Example 1

- Netflix grand prize
 - Movie recommendation system
 - Predicts how a viewer will rate a movie based on his/her ratings
- \$1,000,000 prize!!
(Awarded in 2009)



Motivating Example 2

- Performance modeling
 - How many cashiers do I need in order to keep the average waiting time under 5 minutes?



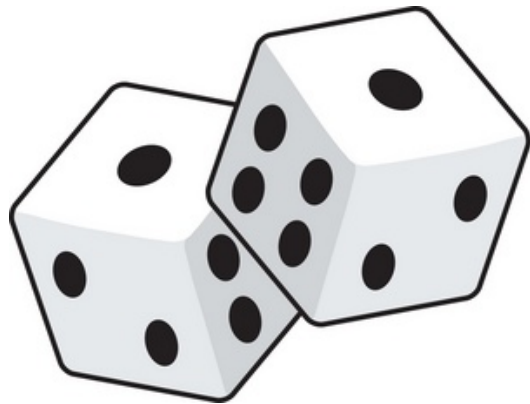
Motivating Example 3

- Classification
 - How does a spam filter know which e-mail is a spam?



Random Experiments

- Outcome varies in an *unpredictable fashion* when the experiment is repeated under the same conditions
- Must specify clearly what to be measured or observed



Experiment 1

Select a ball from an urn containing balls numbered 1 to 50. Note the number of the ball

Possible outcomes:

$$S = \{1, 2, 3, \dots, 50\}$$

$$A = \{1, 3, 5, 7, \dots, 49\}$$

$$P(A) = \frac{25}{50} = \frac{1}{2}$$

Experiment 2

Select a ball from an urn containing balls numbered 1 to 4. Suppose balls 1 and 2 are black and balls 3 and 4 are white. Note the number and color of the ball selected.

Possible outcomes: $\{(1,b), (2,b), (3,w), (4,w)\}$
 $A = \{(3,w), (4,w)\}$
 $P(A) = \frac{2}{4} = \frac{1}{2}$

Experiment 3

Toss a coin 3 times and note a sequence of heads and tails

Possible outcomes: $\{ HHH, HHT, HTT, HTH, THH, THT, TTH, TTT \}$

$$A = \{ HHT, HTH, THH \}$$

$$P(A) = \frac{3}{8}$$

Experiment 4

Toss a coin 3 times and note the number of heads

$A = \{2\}$

Possible outcomes: $\{0, 1, 2, 3\}$

$P(A) = \frac{3}{8}$

Handwritten notes:

- Outcomes with 2 heads: HH T, H T H, T H H
- Outcome with 0 heads: T T T
- Outcome with 1 head: T H T

Experiment 5

A packet is transmitted repeatedly over a noisy channel until it is received without error at the receiver. Count the number of transmissions required.

Possible outcomes: $\{1, 2, 3, \dots\}$

$$P(A) =$$

Experiment 6

Pick a number at random between 0 and 1.

0.1
0.1538
0.0000003

Possible outcomes: $\{x : 0 < x < 1\}$
 $A = \{x : 0 < x < 0.3\}$

$$p(A) = \frac{0.3}{1} = 0.3$$

Experiment 7

Pick two numbers at random between 0 and 1.

Possible outcomes: $\{(x, y) : 0 < x < 1, 0 < y < 1\}$

$$A = x > y = \{(x, y) : 0 < x < y < 1\}$$

$$P(A) = \frac{1}{2}$$

Experiment 8

Pick a number X at random between 0 and 1, and pick a number Y at random between 0 and X .

Possible outcomes:

Sample Space

- Definition: The sample space S of a random experiment is the *set of all possible outcomes*
- Exercise:
 - Define the sample spaces of Exp1 - Exp8

Types of Sample Spaces

- *Discrete* — Countable
 - Outcome can be put into one-to-one correspondence with positive integers.
 - May be finite or countably infinite
- *Continuous* — Uncountable

Events

- *Subset* of the sample space
- Special events
 - Certain event — the sample space itself, consists of all possible outcomes
 - Impossible event — the null event, contains no outcome

Example Events of Exp1 - Exp8

- $E1$ = "An even-numbered ball is selected"
- $E2$ = "The ball is white and even-numbered"
- $E3$ = "The three tosses give the same outcome"
- $E4$ = "The number of heads equal the number of tails"
- $E5$ = "Fewer than 10 transmissions are required"
- $E6$ = "The selected number is non-negative"
- $E7$ = "The two number differs by less than $1/10$ "
- $E8$ = "The two number differs by less than $1/10$ "

Experiments Summary

Experiment	Sample Space	Event
1	$S1 =$	$E1 =$
2	$S2 =$	$E2 =$
3	$S3 =$	$E3 =$
4	$S4 =$	$E4 =$
5	$S5 =$	$E5 =$
6	$S6 =$	$E6 =$
7	$S7 =$	$E7 =$
8	$S8 =$	$E8 =$

Creating Events

- *Elementary events* — events from a discrete sample space which consists of a *single outcome*
- Events can be combined using set operations
- Set operations
 - Union
 - Intersection
 - Complement

Set Arithmetic

- Commutative:
 - $A \cup B = B \cup A$
 - $A \cap B = B \cap A$
- Associative:
 - $(A \cup B) \cup C = A \cup (B \cup C)$
 - $(A \cap B) \cap C = A \cap (B \cap C)$

Set Arithmetic (2)

- Distributive:
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- De Morgan's Rules:
 - $(A \cup B)' = A' \cap B'$
 - $(A \cap B)' = A' \cup B'$

Probability

- Probabilities are numbers assigned to events to indicate how “likely” it is that the events will occur
- A probability law is a rule that assigns probabilities to the events of the experiment
- A probability law is a function that *assigns a number to a set (event)*

Simple Probability Assignment

- If the sample space is discrete with n elements and each outcome is *equally likely*, then we can assign each elementary event a probability of $1/n$

Example 1.1

- An urn contains 10 identical balls numbered 0, 1, ..., 9. Select a ball and note the number on the ball. Find the probability of the following events:
 - $A = \text{"number of ball selected is odd"}$
 - $B = \text{"number of ball selected is multiple of 3"}$
 - $C = \text{"number of ball selected is less than 5"}$
 - $D = \text{"number of ball selected is odd or less than 5"}$

Example 1.2

- Toss a coin 3 times. Find the probability of getting 2 heads out of 3 tosses.

Assigning Probability: Continuous

- Involve assigning numbers to intervals on the real line and region in the plane
- In case of equally likely outcomes, assign probability based on length of interval or size of region in the plane

Example 1.3

- Pick a random number between 0 and 1. Find the probability of the following events.
 - $A =$ “the selected number is less than 0.5”
 - $B =$ “the selected number is between 0.3 and 0.6”
 - $C =$ “the selected number is negative”

Example 1.4

- Pick 2 numbers x and y at random between 0 and 1. Find the probability of the following events
 - $A = "x > 0.5"$
 - $B = "y > 0.5"$
 - $C = "x > y"$

Axioms of Probability

- Axiom 1:

$$0 \leq P(A)$$

- Axiom 2:

$$P(\Omega) = 1$$

Axioms of Probability (2)

- Axiom 3:

If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$

Corollaries

- Corollary 1:

$$P(A') = 1 - P(A)$$

- Corollary 2:

$$P(A) \leq 1$$

- Corollary 3:

$$P(\emptyset) = 0$$

Corollaries (2)

- Corollary 4:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$